Introduction

We know that everything that occupies space and has mass is called matter. Do you know what is matter is composed of? We have studied earlier that matter is composed of tiny little particles, which cannot be seen with naked eye. That particle is called atom. In this lesson, we will study about atoms, molecules, elements, compounds, chemical formulae and atomicity.

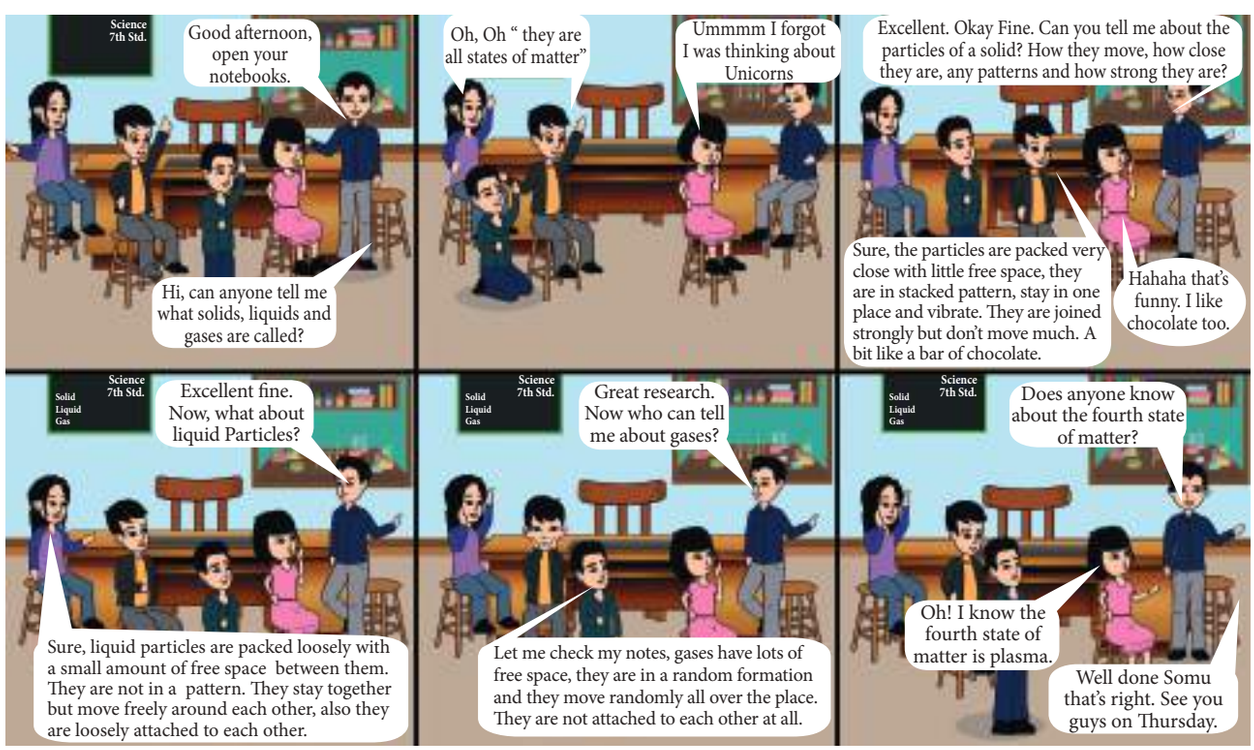
3.1 Atoms

Graphite lead used in pencil is made up of an element called carbon. We can break graphite into smaller and smaller pieces. If we have a finer knife, we can break it even smaller. If we keep cutting the minuscule graphite into smaller and smaller particle, we will reach a point where we get the smallest constituent of graphite - carbon atom. If we break the carbon atom apart, the properties of carbon are exhibited. The smallest unit of an element that exhibits the properties of that element is called as 'atom'. All the matter is composed of tiny particles called atom. Water, rice and everything we see around is made up of atoms. An atom is the basic unit of a matter.



Structure of an atom

Even with the best of optical microscope we cannot see atoms. However, there are advanced instruments that help us to imagine the atoms on the surface of a material. For example, the following figure shows the image of the surface of silicon.



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